Name: ____________________________ Date: ________________ Period: _________

1. For each rational function below, find all holes and vertical asymptotes.

   (a) $r(x) = \dfrac{x^2 - 5x + 6}{x - 3} = \dfrac{(x-3)(x-2)}{x-3}$
   Shared zero $x = 3$, $m = n = 1$ so **Hole**. Simplified: $x - 2$. $L = 3 - 2 = 1$.
   Holes: $(3,\ 1)$ VAs: **none**

   (b) $r(x) = \dfrac{x^2 - x - 12}{(x-4)(x+1)} = \dfrac{(x-4)(x+3)}{(x-4)(x+1)}$
   Shared zero $x = 4$, $m = n = 1$ so **Hole**. Simplified: $\dfrac{x+3}{x+1}$. $L = \dfrac{7}{5}$.
   Holes: $\left(4,\ \dfrac{7}{5}\right)$ VAs: $x = -1$

   (c) $r(x) = \dfrac{x^2 + 3x}{x^2 + 3x + 2} = \dfrac{x(x+3)}{(x+1)(x+2)}$
   No shared zeros (numerator zeros: $0, -3$; denominator zeros: $-1, -2$).
   Holes: **none** VAs: $x = -1$ **and** $x = -2$

2. **Explanation (sample): A hole occurs at a shared zero $x = a$ when the numerator's multiplicity m is greater than or equal to the denominator's multiplicity n ($m \geq n$). After canceling the shared factor, the function still has a finite value at that input. A vertical asymptote occurs when $m < n$ or when the zero appears only in the denominator — the denominator approaches 0 faster than the numerator, causing the outputs to grow without bound.**

3. (a) $\lim_{x \to 3} r(x) = 5$

   (b) **As the input x gets arbitrarily close to 3 (from either side), the output $r(x)$ gets arbitrarily close to 5, even though the point $(3, 5)$ is missing from the graph.**

   (c) Is $r(3)$ defined? **No — The hole means the function is undefined at $x = 3$; it is a removable discontinuity.**

4. $r(x) = \dfrac{x^3 - x^2}{x^2 - x}$

   (a) Numerator: $x^2(x-1)$ Denominator: $x(x-1)$

   (b)

Zero	m	n	Hole or VA?
$x = 0$	**2**	**1**	**Hole** $(m > n)$
$x = 1$	**1**	**1**	**Hole** $(m = n)$

   (c) Simplified: x (cancel $x(x-1)$ from both)
   Holes: $(0,\ 0)$ and $(1,\ 1)$
   (Substitute into simplified form x: at $x = 0$, $L = 0$; at $x = 1$, $L = 1$.)

5. **(AP-Style)** $r(x) = \dfrac{x^2 - 4}{x^2 - x - 2} = \dfrac{(x-2)(x+2)}{(x-2)(x+1)}$

Shared zero $x = 2$, $m = n = 1 \Rightarrow$ **Hole**. Simplified: $\dfrac{x+2}{x+1}$. $L = \dfrac{4}{3}$.

Denominator zero $x = -1$ (not shared) $\Rightarrow$ VA at $x = -1$.

Answer: **A** (Hole at $x = 2$ with $L = \dfrac{4}{3}$; VA at $x = -1$.)